

MODERNISING A LEGACY DATABASE MANAGEMENT SYSTEM

by

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I declare that this dissertation is my own work and that the work of others is acknowledged and indicated by explicit references.

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Abstract

This dissertation explores the process of taking a Legacy system, which is a system that relies on depreciated technology, and making it compatible with more modern technologies. Some of the considerations taken include: ensuring the new system performs the same functions to the same or a higher degree, making sure the actions that the user takes are the same or similar to the original, As well as modernising the system this dissertation also suggests and demonstrates additional features to further improve the user experience of the system, such as various algorithms for providing search recommendations. This will make the system much more accessible to newer users who have less experience with the original system, and possibly improve certain workflows the existing users already take.

Acknowledgements

Template text:

Write any personal words of thanks here. Typically, this space is used to thank your supervisor for their guidance, as well as anyone else who has supported the completion of this dissertation, for example by discussing results and their interpretation or reviewing write ups. It is also usual to acknowledge any financial support received in relation to this work.

Contents

1	Introduction	10
1.1	Background	10
2	Analysis of the legacy system	12
2.1	Technologies used	12
2.2	The current database layout	12
2.2.1	Considerations for the new database	12
2.2.2	Bad database practices	13
3	Design and Implementation	14
3.1	Modern technologies	14
3.2	Database design and Implementation	14
3.3	Website implementation	14
3.4	Testing	15
4	Further development	16
4.1	Features to improve the system further	16
4.2	Search recommendation algorithm design	16
4.2.1	Offline mobile algorithm	16
4.3	Algorithm implmentation and results	17
4.4	Other improvements and their impacts	17

List of Figures

List of Tables

Abbreviations

DBMS Database Management System

BFG Bucks Fungus Group

Chapter 1

Introduction

When it comes to legacy systems many problems can arise if they aren't dealt with in the correct manner, and with the standards of technology constantly changing, and various older technologies becoming outdated or no longer supported, modernising legacy systems is a necessary and very important step to ensuring the longevity of a system. This dissertation aims to demonstrate the process of modernising an existing legacy database management system, which is currently used by an organisation called BFG to keep track of fungi they have found and collected for research purposes, and due to the use of unsupported technology, it's starting to face major problems when meeting new requirements. The steps taken will be, first performing an analysis on the existing database and DBMS to find any issues that need solving and to understand the new requirements of the system. Followed by the design and implementation of the new system with more modern technologies and better practices ensuring it meets all the new requirements. Lastly, I will be implementing 3 variants of the same search recommendation algorithm to demonstrate how database search could be performed whilst foraging for fungi in a place with little to no internet connection.

still to add to the introduction:

- list out the specific techniques covered

1.1 Background

- who bfg are, and what they do + a bit on FRDBI too

- what is the study of mycology and what does it aim to achieve
- a brief description of bfg's database issues

Chapter 2

Analysis of the legacy system

2.1 Technologies used

- description of microsoft access
- the advantages and diadvantages of using microsoft access as a DBMS
- why bfg has to move to something else
- the new requirements that need to be met

2.2 The current database layout

- the format I recieved the data in
- some assumptions made, and why I had to make them

2.2.1 Considerations for the new database

- taxonomy, how the fungi lists need to be formatted to account for this
- the problem with the fungi dictionary being outdated, and why it can't be updated
- lots of unnessassary fields are no longer needed however should be archived still
- new requirements

2.2.2 Bad database practices

- certain fields e.g substrates had predefined lists that weren't always used when creating records, this led to lots of data redundancy
- the format for the fungi dictionary wasn't always followed leading to data redundancy
- format for the site grid refs weren't consistent, making it harder to perform calculations on them
- normalisation problems, some fields had data that was the same as some other fields, e.g a fungus fullName field was just a concatenation of the genus, species, and varity

Chapter 3

Design and Implementation

3.1 Modern technologies

- the reasons for going to a web-based database
- the reason for using django
- how this will help meet the new requirements
- how this will ensure the longevity of the system

3.2 Database design and Implementation

- showing and talking about the erd
- the use of database indexing, and how sharding (partitioning) would also be able to work instead
- in general addressing the considerations listed in chapter 2 and showing how they'll be implemented

3.3 Website implementation

- describe the functionality of the more important webpages

- talking about how the website still keeps the same functionalities whilst making improvements (using comparisons to the original)
- give reason for the improvements/changes to the original
- briefly touch on the security of the website with reference to OWASP
- how parts of the system are scaleable

3.4 Testing

Unsure about this section just yet but I might touch on how I went about making sure the system was up to standards, and show examples of things that were changed based on requirements given to me based on the tests that bfg carried out

Chapter 4

Further development

this section will have additional sections in it if I get round to adding a few more complex ideas

4.1 Features to improve the system further

- discuss the mobile app concept
- the considerations that will be taken into account, the fact its offline ect.

4.2 Search recommendation algorithm design

- give an overview of the alogorithm
- show some maths it uses with justification
- representation of the algorithm in pseudocode
- describe the scalability of the algorithm
- how it's optimised for bigger datasets

4.2.1 Offline mobile algorithm

- overview of the offline mobile variant
- describe the difference and impact on whether or not a few details about the foray are given beforehand

4.3 Algorithm implmentation and results

- compare the different algorithm variants in quality of results and time taken to execute
- also compare based on different dataset sizes

4.4 Other improvements and their impacts

This section will just be about other ways the algorithm could've been improved and aren't actually implemented

- using a model to give suggestions based on images
- more to be added here

Chapter 5

Conclusion and Reflection

- brief summary
- what did I learn
- what is left to be developed before this can be deployed and actually used

Bibliography